

Constitutional AIOps - All Benchmark Tables

Generated: 2026-02-06 14:53:49 UTC Model: constitutional_aiops Source: [benchmark/results/](#)

Table 1: Comprehensive Benchmark Results

Task	Agent	N	Accuracy	BERT-F1	Cos Sim	Term Overlap	P50 (ms)	P95 (ms)
Annotation	qwen3:4b-instruct	100	89.0%	0.516	0.266	0.178	2082	4115
RCA	qwen3:14b	50	90.0%	0.337	0.358	0.649	17096	29403
Overall	Hybrid	150	89.3%	0.456	0.297	0.364	3124	23941

Table 2: Per-Source Breakdown

Task	Source	N	Accuracy	BERT-F1	Cos Sim
Annotation	Loghub Bgl	28	71.4%	0.503	0.223
Annotation	Loghub Hdfs	72	95.8%	0.521	0.283
Rca	Lemma Rca Cloud	28	100.0%	0.349	0.463
Rca	Opseval 5G Communication	2	50.0%	0.294	0.221
Rca	Opseval Mobile Communication Network	4	100.0%	0.352	0.243
Rca	Opseval Wired Network	16	75.0%	0.318	0.222

Table 3: Error Analysis

Failure Mode	Count	Description
BGL False Positive	8	ANN_118, ANN_142, ANN_141, ANN_135, ANN_119 (+3 more)
RCA Incorrect (opseval_Wired Network)	4	RCA_002, RCA_040, RCA_039, RCA_046
ANNOTATION Incorrect (loghub_hdfs)	3	ANN_049, ANN_034, ANN_012
RCA Incorrect (opseval_5G Communication)	1	RCA_067
Total Failures	16	16/150 = 10.7% error rate

Table 4: Ablation Study - Component Contributions

Configuration	Ann Acc	RCA Acc	Overall	BERT- F1	Cos Sim	Term Ov.	Avg Latency	Delta vs Full
Full System (Hybrid dual-agent baseline)	89.0%	88.0%	88.7%	0.458	0.302	0.371	7208ms	-
Single Agent - Qwen3-4B for both annotation and RCA	89.0%	94.0%	90.7%	0.458	0.303	0.389	7250ms	+2.0%
Single Agent - Qwen3-14B for both annotation and RCA	89.0%	92.0%	90.0%	0.457	0.302	0.392	7582ms	+1.3%
No Structured Output - raw text, no JSON metadata extraction	89.0%	92.0%	90.0%	0.458	0.303	0.385	7467ms	+1.3%

N=150 per configuration. Dataset: curated_150 (seed=42). Temperature=0.0.